SAYAK ROY

Kolkata, West Bengal, India
8509331983 • Email: 08sayakroy@gmail.com • LinkedIn: www.linkedin.com/in/sayak-roy • GitHub: github.com/sayakroy83

Summary

Computer Science graduate and full-stack developer with hands-on experience building web applications and RESTful APIs using React, Node.js, Express and MongoDB. Built a learning management system with role-based authentication and an AI chatbot powered by the OpenAI API. Solid foundation in data structures, OOP in Java and database concepts, with a growing interest in cloud technologies and scalable backend systems. Seeking a Software Engineer role to build reliable, user-focused products.

Education

University of Engineering and Management **Aug. 2021 – June 2025**
B.Tech in Computer Science and Technology, CGPA: 7.62 *Kolkata, West Bengal*

Technical Skills

Programming : Java(Proficient), Python(Intermediate), JavaScript
Software Development : Data Structures & Algorithms, OOPS, REST APIs
Backend : Node.js, Express.js
Frontend : React, HTML, CSS, Tailwind CSS
Databases : MongoDB, SQL
Tools : Git, GitHub, Postman, Google Workspace, AWS(Cloud Fundamentals)

Certifications

- **AWS Cloud Technical Essentials – AWS** **Dec 2024**
- **Introduction to Data Analytics – IBM** **Sep 2023**

Experience

Virtual Internship – Data Analytics Intern **June 2024 – Aug 2024**
IBM SkillsBuild × CSRBOX Foundation

- Analyzed real-world datasets using Python (Pandas, NumPy) to identify trends and generate actionable insights.
- Prepared reports demonstrating data-driven decision making.
- Gained hands-on exposure to data-driven decision-making and teamwork in virtual internship setting.

Projects

Learning Management System | *MERN Stack* | **May 2026**

- Designed and implemented RESTful APIs using Node.js and Express.js for authentication, course management, and role-based access control.
- Designed MongoDB data models for users, courses, and enrollments to support scalable data retrieval.
- Implemented backend APIs with a React frontend to deliver complete end-to-end functionality.
- Applied CURD operations while following modular backend architecture.

QuickGPT | *React, Node.js, OpenAI API, Tailwind CSS* | **Jan 2026**

- Built a full-stack AI chatbot integrating the OpenAI API to generate context-aware responses.
- Implemented real-time conversation handling and responsive UI using React and Tailwind CSS.
- Managed asynchronous API requests and frontend state for smooth user interaction.

Crop Prediction & Disease Detection Platform | *Python, React, HTML, Tailwind CSS* | **Nov 2024**

- Collaborated with a 3-member team to develop a smart agriculture platform for crop recommendation, fertilizer prediction, and plant disease detection.
- Developed the React-based frontend, creating responsive user interfaces for crop prediction, fertilizer recommendation, and disease detection workflows.
- Contributed to backend integration by connecting the frontend with Flask APIs to enable real-time prediction results.
- Worked with teammates to test, debug and integrate the complete application, which achieved approximately 90% accuracy in plant disease classification using CNN models.

Languages

- English | Hindi | Bengali